STAT 131: Lecture Notes 5

Winter 2024

5.1 Bivariate distributions [DS 3.4]

So far, we have worked with a single random variable at a time. In many cases we will have two (or more) random variables that we deal with together. For example, a doctor may measure both a person's height and weight, or a company might record both the number of hires and fires in a given year.

Definition 5.1 (Joint distribution)**.** The joint distribution of two random variables X and Y is defined as the collection of all probabilities of the form $\mathbb{P}((X,Y) \in E)$ for all $E \subset \mathbb{R}^2$ such that $\{(X,Y) \in E\}$ is an event.

Definition 5.2 (Discrete joint distribution)**.** A discrete joint distribution is the joint distribution of two random variables X and Y where both X and Y are discrete.

Example 5.1. Outcomes of soccer/basketball games; ♢

Definition 5.3 (Joint pmf)**.** The joint probability mass function (pmf) of two discrete random variables X and Y is defined as the function f such that

$$f(x,y) = \mathbb{P}(X = x \& Y = y).$$

for all $(x,y) \in \mathbb{R}^2$. This may equivalently be written as $\mathbb{P}(X = x, Y = y)$ or $\mathbb{P}(X = x \cap Y = y)$.

Note that $f(x,y) \geq 0$ for all $(x,y) \in \mathbb{R}^2$ and $\sum_x \sum_y f(x,y) = 1$.

Example 5.2 (auto insurance)**.** Suppose we have the following joint pmf:

Stolen	Brand 1	Brand 2	Brand 3	Brand 4	Brand 5	Total
No	0.129	0.298	0.161	0.280	0.108	0.976
Yes	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

♢

Example 5.3. Use the below joint pmf to answer the following questions:

	$Y=1$	$Y=2$	$Y=3$
$X=0$	1/8	1/4	1/8
$X=1$	1/8	0	3/8

1. What is $\mathbb{P}(X=0, Y=1)$?
2. What is $\mathbb{P}(X=0)$?
3. What is $\mathbb{P}(Y=1)$?

1) $P(X=0, Y=1) = 1/8$

2) $P(X=0) = \sum_{y=1}^{3} P(X=0|Y=y)\,P(Y=y) = \sum_{y=1}^{3} P(X=0, Y=y) = 1/8 + 1/4 + 1/8 = 1/2$

3) $P(Y=1) = \sum_{x=0}^{1} P(X=x, Y=1) = 1/8 + 1/8 = 1/4$

♢

Definition 5.4 (Continuous joint distribution)**.** A continuous joint distribution is the joint distribution of two random variables X and Y where both X and Y are continuous.

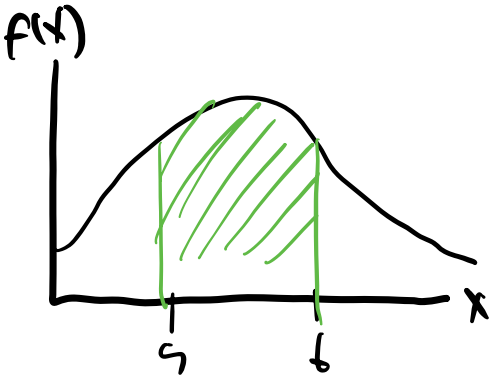
Example 5.4. Stock prices of Google and Apple, precipitation and wind speed, human height and weight, etc.. ♢

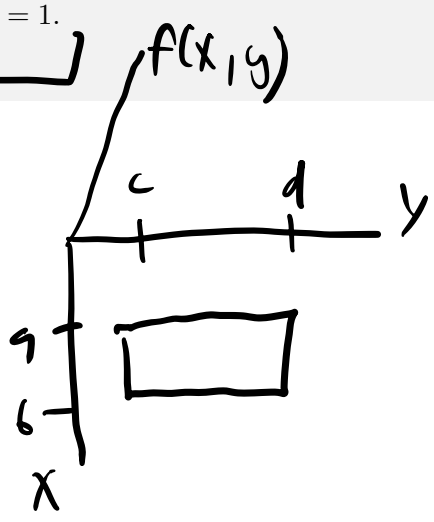
Definition 5.5 (Joint pdf)**.** The joint probability density function (pdf) of two continuous random variables X and Y is the non-negative function $f(x,y)$, where

$$\mathbb{P}(a \le X \le b, c \le Y \le d) = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

Note that

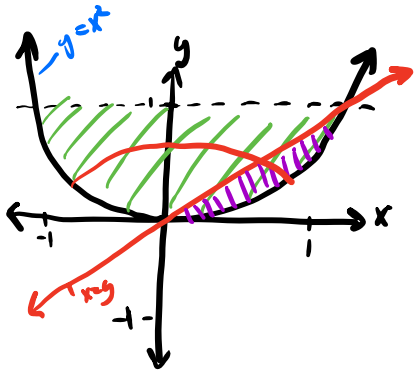
$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1.$$





Example 5.5. Suppose the joint pdf of (X, Y) is

$$f(x,y) = \begin{cases} cx^2y, & \text{for } x^2 \le y < 1 \\ 0, & \text{otherwise} \end{cases}$$



- Find c.
- Calculate $\mathbb{P}(X > Y)$.
- Calculate $\mathbb{P}(X = Y)$.

1) $1 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dy\,dx = \int_{-1}^{1}\int_{x^2}^{1} cx^2y\,dy\,dx$

$= \int_{-1}^{1}\left[\frac{c}{2}x^2y^2\Big|_{x^2}^{1}\right]dx$

$= \int_{-1}^{1}\frac{c}{2}(x^2-x^6)\,dx = \frac{c}{6}x^3 - \frac{c}{14}x^7\Big|_{-1}^{1} = \left(\frac{c}{6}-\frac{c}{14}\right)-\left(-\frac{c}{6}+\frac{c}{14}\right) = \frac{4c}{21}$

$\Rightarrow c = 21/4$

2) $P(X>Y) = \int_0^1\int_{x^2}^{x}\frac{21}{4}x^2y\,dy\,dx = \int_0^1\left[\frac{21}{8}x^2y^2\Big|_{x^2}^{x}\right]dx$

$= \int_0^1\frac{21}{8}(x^4-x^6)\,dx = \frac{21}{40}x^5 - \frac{21}{56}x^7\Big|_0^1 = \frac{21}{40}-\frac{21}{56} = \boxed{3/20}$

3) $P(X=Y) = 0$

The line $X=y$ has no area!

$F(x) = P(X \leq x)$ ♢

Definition 5.6 (Joint CDF)**.** The joint cumulative distribution function (cdf) of two random variables X and Y is the function F such that for all $(x, y) \in \mathbb{R}^2$,

$$F(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

Theorem 5.1. *For two continuous random variables X and Y, the following relationship exists between the joint cdf and the joint pdf:*

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = f(x, y).$$

Note that $\frac{\partial^2}{\partial x \partial y} g(\cdot)$ is notation that means we take the derivative of $g(\cdot)$ with respect to the variable y, and then we take a derivative again, with respect to x. We can also change the order of differentiation.

Example 5.6. $F(x, y) = \frac{1}{16} xy(x + y)$ for $x \in [0, 2]$ and $y \in [0, 2]$. Find joint pdf.

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

$$\frac{\partial}{\partial x} \frac{1}{16} xy(x+y) = \frac{\partial}{\partial x} \frac{1}{16} (x^2 y + y^2 x) = \frac{1}{16} (2xy + y^2)$$

$$\frac{\partial}{\partial y} \frac{1}{16} (2xy + y^2) = \frac{1}{16} (2x + 2y) = f(x, y)$$

for $x \in [0,2]$ and $y \in [0,2]$ ♢

check that this integrates to 1

5.2 Marginal Distributions [DS 3.5]

The joint pmf/pdf tells us how two variables X and Y vary jointly. However, sometimes we just want to know how one variable (say X) varies, *regardless* of the other variable Y. In other words we are interested in the marginal distribution, which is just the distribution of a single random variable. We can use the joint pmf/pdf to find this marginal pmf/pdf.

Theorem 5.2. *For two discrete random variables X and Y, with joint pmf $f(x,y)$, the marginal pmf of X is*

$$f_X(x) = \sum_y f(x,y).$$

Note, we use the notation $f_X(x)$ just to clarify that the marginal distribution is for the variable X. We may also use $f_Y(y)$ to denote the marginal distribution of Y.

Proof. Use the law of total probability.

$$f_X(x) = P(X=x) = \sum_y P(X=x \mid Y=y)\, P(Y=y) = \sum_y P(X=x \cap Y=y)$$

$$= \sum_y f(x,y)$$

□

Theorem 5.3. *For two continuous random variables X and Y, with joint pdf $f(x,y)$, the marginal pdf of X is*

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy.$$

Similarly, the marginal pdf of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx.$$

Example 5.7. For two continuous random variables X and Y, let

$$f(x,y)\begin{cases} x+y, & \text{for } 0\le x\le 1,\ 0\le y\le 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find the marginal density of X.

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)\,dy=\int_{y=0}^{y=1} x+y\,dy = xy+\tfrac{1}{2}y^2\Big|_0^1 = \boxed{x+1/2}$$

for $0\le x\le 1$

0, otherwise

♢

5.3 Independent Random Variables [DS 3.6]

Recall that for a pair of events A and B, they are independent if and only if

$$P(A\cap B)=P(A)P(B).$$

We will now extend the concept of independence to random variables.

Definition 5.7 (Independence of two random variables)**.** We say that two random variables X and Y are independent, denoted as $X\perp Y$, if and only if

$$f(x,y)=f_X(x)f_Y(y).$$

Note that this is true for both discrete random variables (pmfs) and continuous random variables (pdfs).

Example 5.8. For two continuous random variables X and Y, let

$$f(x,y)\begin{cases} x+y, & \text{for } 0\le x\le 1,\ 0\le y\le 1 \\ 0, & \text{otherwise} \end{cases}.$$

Is $X\perp Y$?.

$f_X(x)=x+1/2$ $\qquad$ $f_Y(y)=\int_0^1 x+y\,dx = y+1/2$

$f_X(x)\cdot f_Y(y)=(x+1/2)(y+1/2)=xy+\tfrac{1}{2}x+\tfrac{1}{2}y+\tfrac{1}{4}\neq x+y=f(x,y)$

So X is not indep. of Y.

♢

Theorem 5.4 (Factorization). *If the joint pdf of X and Y satisfies*

$$f_{X,Y}(x,y) = h(x)g(y)$$

for all $x, y \in \mathbb{R}$, where $h(\cdot)$ and $g(\cdot)$ are two univariate functions, then $X \perp Y$.

Special case: let $h_X(x) = f_X(x)$ and $h_Y(y) = f_Y(y)$, so

$$f(x,y) = f_X(x) f_Y(y).$$

Example 5.9. For two continuous random variables X and Y, let

$$f(x,y) \begin{cases} 4xy, & \text{for } 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}.$$

Is $X \perp Y$?.

$$f_X(x) = \int_0^1 4xy\,dy = 2xy^2\Big|_0^1 = 2x$$

$$f_Y(y) = \int_0^1 4xy\,dx = 2y$$

$$f_X(x) \cdot f_Y(y) = 2x \cdot 2y = 4xy = f(x,y)$$

$$\Rightarrow X \perp Y$$

$h(x) = 4x$

$g(y) = y$

or

$$f(x,y) = 4xy = 4x \cdot y = h(x)\,g(y)$$

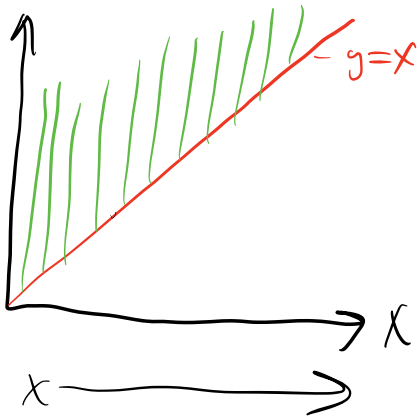
$$\Rightarrow X \perp Y$$

♢

Note that if the support of one variable depends on the other, then they cannot be indep.

$$X \perp Y \Rightarrow f(x,y) = f_X(x) \cdot f_Y(y)$$

Example 5.10. Suppose that the time until server 1 crashes is denoted by $X \sim \text{Exp}(\lambda)$ and the time until server 2 crashes is denoted by $Y \sim \text{Exp}(\mu)$. Assume that $X \perp Y$. What is the probability that server 1 crashes before server 2?



$$P(X < Y) = \int_0^\infty \int_x^\infty f(x,y)\, dy\, dx$$

$$= \int_0^\infty \int_x^\infty \lambda e^{-\lambda x} \mu e^{-\mu y}\, dy\, dx$$

$$= \int_0^\infty \lambda e^{-\lambda x} \int_x^\infty \mu e^{-\mu y}\, dy\, dx$$

$$= \int_0^\infty \lambda e^{-\lambda x} e^{-\mu x}\, dx = \lambda \int_0^\infty e^{-(\lambda+\mu)x}\, dx$$

$$= \lambda \left[\frac{1}{\lambda+\mu} e^{-(\lambda+\mu)x} \Big|_0^\infty \right] = \boxed{\frac{\lambda}{\lambda+\mu}}$$

♢

Theorem 5.5. *If X and Y are independent, then $g(X)$ and $h(Y)$ are independent for any function g and h.*

Given the following joint pmf

$$f(x,y) = \begin{cases} \frac{x^2+y^2}{31} & \text{for } x=1,2,3 \text{ and } y=0,1 \\ 0 & \text{otherwise} \end{cases}$$

1) Find $f_X(x)$
2) Find $f_Y(y)$
3) Is $X \perp Y$?

1) $f_X(x) = \sum_y f(x,y) = \frac{x^2+0^2}{31} + \frac{x^2+1^2}{31} = \frac{2x^2+1}{31}$ for $x=1,2,3$

2) $f_Y(y) = \sum_x f(x,y) = \frac{1^2+y^2}{31} + \frac{2^2+y^2}{31} + \frac{3^2+y^2}{31} = \frac{14+3y^2}{31}$ for $y=0,1$

3) $f_X(x) \cdot f_Y(y) = \frac{2x^2+1}{31} \cdot \frac{14+3y^2}{31} \neq f(x,y) = \frac{x^2+y^2}{31}$

So X and Y are <u>not</u> indep.